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The influence of existing interventions on sleep of youth with attention-deficit/hyperactivity disorder: A meta-analysis of randomized controlled trials☆

Yuan Fang [a,1], Zixin Wang [b,1], Mengge Zhang [c], Fenghua Sun [a,*]

[a] Department of Health and Physical Education, The Education University of Hong Kong, Hong Kong Special Administrative Region of China
[b] Centre for Health Behaviours Research, JC School of Public Health and Primary Care, The Chinese University of Hong Kong, Hong Kong Special Administrative Region of China
[c] Department of Educational Psychology, Faculty of Education, The Chinese University of Hong Kong, Hong Kong Special Administrative Region of China

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ABSTRACT

Sleep problems are commonly observed among children and adolescents with attention-deficit/hyperactivity disorders (ADHD). We completed a literature search by 2025-6-9 on EBSCOhost, PubMed, Web of Science, and Scopus to quantify the effects of any existing interventions on sleep outcomes among children/adolescents with ADHD. From a pool of 4081 studies, 40 randomized controlled trials (RCTs; 75% with low-risk-of-bias, 4361 participants and 65.2% attended interventions) were eligible for a pairwise meta-analysis (PWMA). Of these, 28 RCTs were included in a Bayesian network meta-analysis (NMA). Results were yielded from both mean change and post-test measurement of the outcomes if available. In PWMA (moderate-to-high certainty), the examined interventions had small and significant effects on sleep disturbance (standardized mean difference [SMD] by mean change [95% CI]: overall [k = 29]: −0.30 [-0.42, −0.17], $p < 0.001$; sleeping pills [k = 7]: −0.10 [-0.41, 0.22], $p > 0.05$; ADHD drugs [k = 6]: −0.25 [-0.52, 0.01], $p > 0.05$; behavioral sleep [BS] interventions [k = 10]: −0.48 [-0.70, −0.27], $p < 0.001$) and sleep onset latency (overall [k = 12]: −0.41 [-0.67, −0.14], $p < 0.01$; sleeping pills [k = 3]: −0.68 [-1.08, −0.27], $p < 0.01$; ADHD drug [k = 1]: −0.39 [-0.96, 0.18], $p > 0.05$; BS interventions [k = 5]: −0.36 [-0.88, 0.16], $p > 0.05$). The examined interventions did not have significant effects on sleep efficacy, sleep duration, latency to persistent sleep or daytime sleepiness. Results of NMA showed that sleeping pills had the best performance in shortening SOL, while BS interventions may be the best choice for reducing sleep disturbance. However, more rigorous evidence is warranted. This review indicates that a multi-component and tailored-designed intervention represents important personalized strategies for improving sleep in youths with ADHD.

Glossary of terms

ADHD (in main text)	attention-deficit/hyperactivity disorder
ADHD (in Fig. S3)	ADHD drugs
ASPD	advanced sleep phase disorder
BPT	behavioral parent training
BS	behavioral sleep (intervention)
BSE	behavioral sleep education
BST	behavioral sleep training
CBT	cognitive behavioral therapy
CDR108	*Bacopa monnieri*
CG	control group
CHSQ	children's sleep habits questionnaire
CI	confidence intervals
CPA	cognitive training-combined/engaged physical activity
CRSD	circadian rhythm sleep disorder
Cyp	cyproheptadine

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* Corresponding author.Rm D4-2/F-25, The Education University of Hong Kong, No. 10 Lo Ping Road, Tai Po, NT, Hong Kong Special Administrative Region of China.
E-mail addresses: lunajoef@gmail.com (Y. Fang), wangzx@cuhk.edu.hk (Z. Wang), miawork9090@gmail.com (M. Zhang), fhsun@eduhk.hk (F. Sun).

[1] These authors contributed equally to this work.

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(*continued*)

D1/D2	dopamine receptor
DAT	dopamine transporter
DAT1	dopamine transporter gene, also known as SLC6A3
DIC	deviance information criterion
EEG	electroencephalography
ER	extended release
ES	effect size
GABA	the brain chemicals gamma aminobutyric acid
GRADE	the grading of recommendation assessment, development, and evaluation
GXR	extended-release-guanfacine
HD	high dose
IG	intervention group
IR	immediate release
IRB	internal review board
LD	low dose
LPS	latency to persistent sleep
MAS	mixed amphetamine salts
MCMC	Markov chain Monte Carlo
MD (in main text)	mean difference
MD (in Table S3)	moderate dose
MPH	methylphenidate
MTS	methylphenidate transdermal system
NAASO	number of awakenings after sleep onset
NMA	network meta-analysis
Non-pharm/NPh	non-pharmaceutical interventions
NREM	non-rapid eye movement
OR	odds ratio
OROS	osmotic-release oral system
OSA	obstructive sleep apnea
PLMS	periodic limb movements in sleep
PRISMA	preferred reporting items for systematic reviews and meta-analyses
PREMB-R PM	at-home functioning at late-afternoon and evening
PSPSI	paediatric sleep problem survey instrument
PSQI	Pittsburgh sleep quality index
PST	parent sleep training
PWMA	pairwise meta-analysis
RCT	randomized controlled trials
REM	rapid eye movement
RLS	restless legs syndrome
RoB 2	risk of bias 2
SD	standard deviation
SDB	sleep-disordered breathing
SDQ	sleep disorders questionnaire
SE	sleeping efficacy
SMD	standardized mean difference
SOL	sleep onset latency
so-tDCS	slow oscillating transcranial direct current stimulation
SP	sleeping pills
SUCRA	surface under the cumulative ranking curve
TST	total sleeping time
UC	usual care
US	United States of America
WASO	wake time after sleep onset

1. Introduction

Attention-deficit/hyperactivity disorder (ADHD) is a commonly diagnosed developmental disorder [1] with a 5.9–7.1 % worldwide prevalence in children and adolescents [2,3]. Besides well-known symptoms such as inattention and/or hyperactivity/impulsiveness [2], sleep problems are also commonly observed in individuals with ADHD. Approximately 70% of children with ADHD present different sleep problems [4,5], and these children have experienced more sleep problems than those without this condition [6–8]. These problems [7–9] have been detected using subjectively measured variables such as bedtime resistance, sleep-onset difficulties, night awakening, difficulties in awakening in the morning, sleep-disordered breathing (SDB), and daytime sleepiness, as well as using objectively measured variables (e.g., sleep onset latency [SOL] and latency to persistent sleep [LPS]).

Based on the existing evidence, a bidirectional relationship has been established between ADHD symptoms and sleep problems [10,11]. It proposes that sleep problems may be an early predictor of ADHD onset [11,12] and may be negatively associated with poorer child psychosocial situations [13]. Several associated factors have been identified in the literature. First, children with ADHD were found to have a reduced number of cyclic alternating pattern (CAP) sequences and a lower CAP rate than normal controls, which was due to the low CAP rate during sleep stage 2 non-rapid eye movement (NREM) sleep [14]. Consistent to this, ADHD population were reported to have a reduction in slow-wave activity across the whole brain, reduced delta power in deep sleep but increased delta/theta activity when they were compared with normal controls [15,16]. These abnormal brain activity during sleep may further aggravate cognitive dysfunction in those with ADHD [15–17]. Second, several neurobiological mechanisms are embedded in sleep problems associated with ADHD, including delayed circadian rhythms [12,18], dopaminergic dysfunction [19], and commencement of melatonin synthesis [20–22]. Third, sleep disturbances may also correlate with problems in brain morphology and network connectivity in ADHD [23], such as smaller gray matter volumes in the middle and inferior frontal gyri, amygdala, striatum, and insula. Fourth, sleep disorders (such as restless legs syndrome [RLS], periodic limb movements [PLMS], obstructive sleep apnea [OSA]) are observed to associated with ADHD [24–26]. Studies show a high prevalence of PLMS and RLS in individuals with ADHD, with some evidence linking iron deficiency to both ADHD and RLS pathophysiology [27]. As a result, many ADHD symptoms resemble those of sleep deprivation, and children and adolescents with ADHD who experience a lack of sleep may also exhibit hyperactivity and impulsivity [28].

So far, it has been revealed that current drugs for treating ADHD (e.g., stimulant) may cause adverse events such as decreased appetite or even worsening of sleep problems especially if taken later in the day [29]. ADHD medication may interfere the physiological arousal level in those affected by ADHD. Thus, sleep problems in children/adolescents with ADHD can be caused by ADHD drugs [30] or ADHD per se [31,32]. Therefore, the reduction of sleep problems and enhancement of sleep health play important roles in the ADHD management, neuro-development, and well-being of children and adolescents with ADHD.

Several types of interventions have been applied to treat sleep problems among children and adolescents with ADHD [9,33], including pharmaceutical treatments (e.g., melatonin, which can regulate circadian rhythms and inhibit presynaptic dopamine release [34]) and non-pharmaceutical interventions (e.g., behavioral sleep training and sleep hygiene education). Although some sleeping pills [35] and behavioral therapy [9] have been reported to be effective in treating insomnia, a high relapse rate of sleep-onset insomnia when resuming melatonin treatment was also reported in a long-term follow-up study [36]. Thus, the identification of efficient and sustainable strategies is necessary to help children and adolescents with ADHD improve the quality of sleep.

Therefore, a comprehensive systematic review and meta-analysis of the effects of existing interventions on sleep of children and adolescents with ADHD is warranted; Yet, it is not available. To fill this knowledge gap, the present study aimed to (1) quantify the extent to which the existing interventions influence the sleep of children and adolescents with ADHD, (2) compare the intervention effects on sleep according to various intervention types and subtypes, and (3) calculate the relative effect of existing interventions on participants' sleep and rank their probable performance.

2. Methods

This review was registered in PROSPERO (Ref No.: CRD420251070940). As this review was conducted and analyzed based on published data, internal review board (IRB) approval was not required.

2.1. Search strategy

Relevant journal articles were identified by searching PubMed, Web of Science, EBSCOhost, and Scopus databases and platforms covering the published periods of 1966–2025, 1975–2025, 1929–2025, and 1970–2025, respectively; the search was completed on June 9, 2025 (Fig. 1). The Boolean operator was used in the search strategy with "OR" and/or "AND" used to link search terms. The asterisk "*" was used as a wildcard symbol appended at the end of the terms to search for variations of those terms. The complete and detailed search strategies for the title, abstract, or topic are presented in Table 1 and Table S1, respectively.

Table 1
Search strategy.

No	Item	Terms
1	ADHD	ADHD OR attention deficit-hyperactivity disorder OR attention deficit hyperactivity disorder OR attention deficit/hyperactivity disorder OR attention-deficit/hyperactivity disorder
2	Children or adolescents	children or adolescents OR youth or child OR teenager OR pediatric OR paediatric OR kids OR students
3	RCT	RCT OR randomized control trial OR randomized controlled trial OR randomised control trial OR randomised controlled trial OR controlled trial OR experimental design OR experiment
4	Research question	1 AND 2 AND 3
5	Handsearching	Checking through the reference lists of the relevant published reviews

2.2. Inclusion and exclusion criteria

Inclusion criteria for the review were studies having to (1) be a true-experiment (i.e., a randomized controlled trial, RCT), (2) be conducted in children or adolescents (age 0-19 years old) diagnosed with ADHD; (3) report the effect of an intervention for children and/or adolescents with ADHD (e.g., the effect of a sleep medicine or a sleep health education program); (4) report the sleep-related continuous variables of children and/or adolescents with ADHD (e.g., mean and standard deviation [SD]); (5) use between-group comparisons to evaluate the effectiveness of the interventions (i.e., comparing the effect between intervention and control group); (6) be published in English, peer-reviewed journals; and (7) have an available full text.

An article was excluded if (1) participants were adults; (2) it focused on the outcomes of parents, teachers, or health professionals; (3) it had a

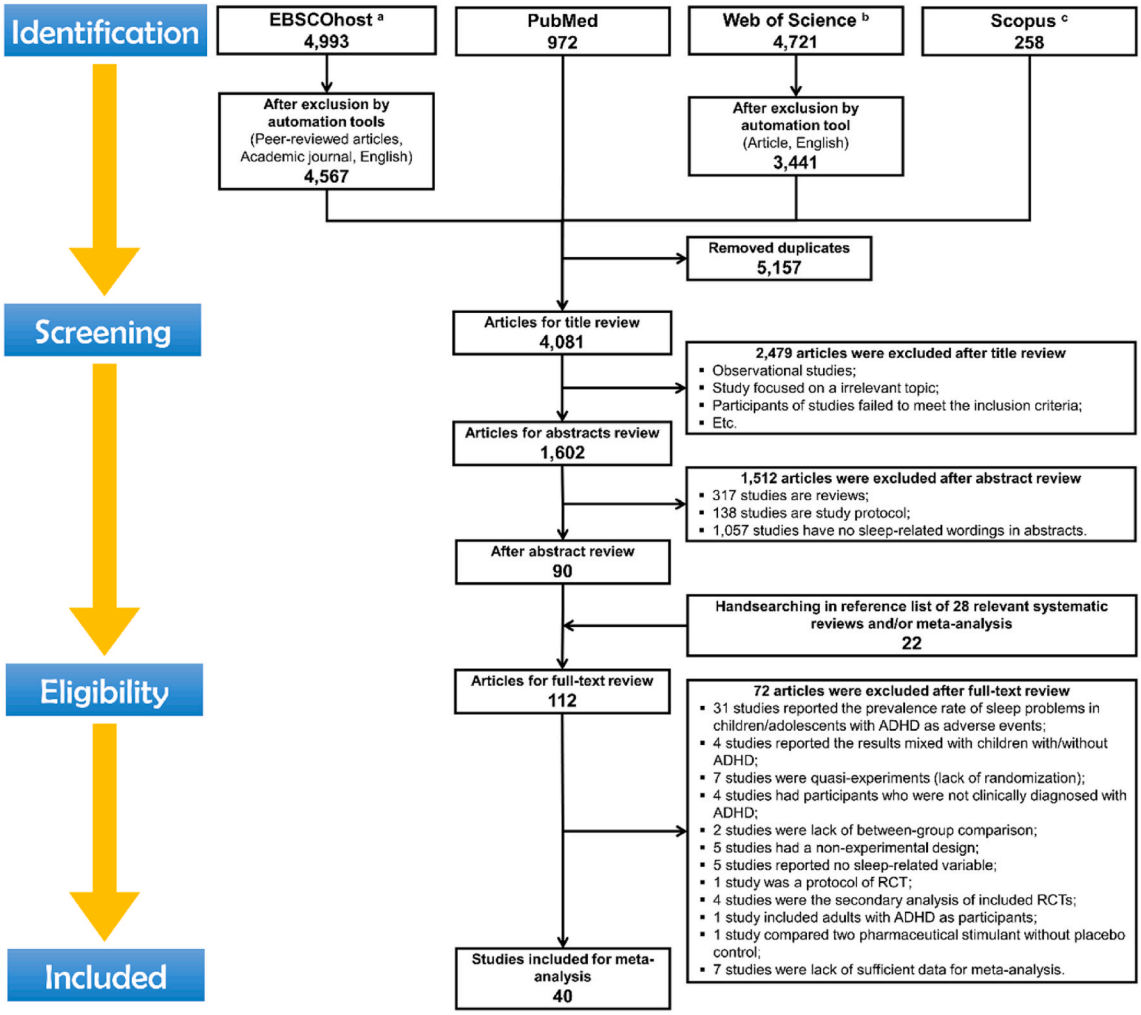


Fig. 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram: Searching process for studies selection about the RCT-assessed interventions (with report of sleep outcomes) among children and adolescents with ADHD.
Note. a Included databases of EBSCOhost are Academic Search Ultimate, APA PsycArticles, APA PsycInfo, CINAHL with Full Text, ERIC, MEDLINE, SPORTDiscus with Full Text, British Education Index, Education Source Ultimate, Library, Information Science & Technology Abstracts with Full Text, Primary Search; b Included databases of Web of Science are A&HCI, ESCI, CPCI-SSH, CPCI-S, SCI-EXPANDED, and SSCI; c Included publications of Scopus are both final published and accepted in press.

quasi-experimental or non-experimental study design; (4) it had no report on sleep-related outcomes of children and/or adolescents with ADHD; (5) it reported non-continuous sleep-related variables; (6) it focused on the prevalence of insomnia in children and/or adolescents with ADHD that was reported as safety data of adverse events; (7) it used animal models; (8) it described the development/validation of a scale/tool; (9) it focused on modelling or projection; (10) it was a study protocol, a review, an editorial, a commentary, a discursive paper, a conference abstract, or an unpublished dissertation; or (11) the full-text form was not available.

2.3. Screening process

Two authors (YF and ZW) were involved in screening of articles, and a consensus by comparison or discussion was reached between the authors during the process. As shown in Fig. 1, after initial execution of the search strategy and exclusion using automated tools in each database, the remaining papers were combined and checked for duplicates. After removing duplicates, 4081 articles were screened for relevance by reviewing their titles. Then the abstracts of 1602 articles were reviewed for eligibility, yielding 90 articles for full-text screening. In addition, 22 papers were included after checking the reference lists of 27 previously published systematic reviews in relevant research areas (listed in Table S2). Overall, 112 papers were read in full to determine whether they met the inclusion criteria. Finally, 40 articles were included in the meta-analysis. The inter-rater reliability analyses showed that Cohen's kappa ranged from 0.941 to 0.956 during the article screening process, which indicates almost perfect agreement between two reviewers (see Table S3). These studies provided sufficient data to compare the effect sizes of any primary endpoints, that is, sleep duration (i.e., objectively-measured total sleeping time [TST] and self-reported data), sleeping efficacy (SE), SOL, sleep disturbance, LPS, and daytime sleepiness. Since the majority of included studies provided post-outcome measures (38/40) and had a balanced baseline (36/40), the post-test measurements were selected as endpoint outcomes for meta-analysis.

2.4. Assessment for publication quality and risk of bias

The risk of bias in the included articles was assessed by risk of bias 2 (RoB 2) in RCTs [37]. The results were categorized as low risk, some concerns, or high risk (Table S4**)**. Each item was assessed by the same authors who had completed the review/screening process, and a consensus was reached through comparison or discussion when necessary.

2.5. Data extraction

Relevant information from the included articles was analyzed and summarized using a standardized table with categorized themes of key information (Table S5). This table includes the publication year, country, characteristics of participants, intervention content in both the intervention and control groups, and the effectiveness of the designed interventions in terms of the measures utilized and the key results. Two authors (as indicated above) independently completed the review/screening process and data extraction, with a consensus regarding the final study inclusion, analyses, and summary was achieved via discussion.

2.6. Pairwise meta-analysis

RevMan v.5.4.1 (The Cochrane Collaboration, Software Update, Oxford, UK) and Stata v.16 (StataCorp LLC, Texas, US) were used to conduct a pairwise meta-analysis (PWMA). If the mean or SD of an outcome was not provided, it was estimated from the corresponding median, standard error (SE), range, and number of participants in each arm, based on the formula described in the Cochrane handbook [38]. The extracted post-test outcomes (both parallel- and crossover-arm RCTs) and/or mean change of pre-/post-test (for parallel-arm RCTs only, since crossover-arm RCTs had no insufficient data, see Appendix 1) were operationalized as continuous measures for comparison, which may have resulted in a high level of heterogeneity between different studies. Thus, SMD (calculated as Hedge's adjusted *g* with adjustment for small sample bias [39], Appendix 1) with 95% confidence intervals (CI) was calculated using a random-effects model. Statistical significance was set at $p < 0.05$. The SMD (95% CI) was perceived as trivial (0.00–0.19), small (0.20–0.49), medium (0.50–0.79), and large (≥0.80) [40,41]. Subgroup analyses were conducted to detect the potential effect modifiers by using a random-effects model. The potential effect modifiers were primarily grouped by (1) pharmacological species for sleeping pills (melatonin, Z-drugs, L-Theanine, L-Dopa, and antihistamines) and ADHD drugs (stimulant vs non-stimulant), and (2) intervention content or intensity for non-pharmaceutical interventions (BS vs non-BS, or Hiscock protocol vs other more-intensive-scheduled protocols). For a meta-analysis of a small number of included studies, the *Q* statistic test, a comprehensive test of the homogeneity of effect sizes and the impact of moderators, was applied [42]. Furthermore, heterogeneity (%) was assessed by the I^2 statistic, and the I^2 value was interpreted as small (<25%), medium (25-75%), or large (>75%), as per convention [43]. The PWMA results are presented in Table 2 and Table S6.

2.7. Assessment of publication bias

The potential publication bias of the selected endpoints was presented using funnel plots with SMD on the x-axis and the standard error of SMD on the y-axis. All endpoints were further assessed using Egger's test to detect asymmetry of the funnel plots [44]. Moreover, the trim-and-fill method was used to estimate the true effect sizes, adjusted for potential publication bias and the number of studies needed to balance the funnel plot [45]. Significance was set at $p < 0.05$ for all tests. The results are summarized in Table S7.

2.8. Evidence grading of traditional meta-analytic results

For PWMA results, the certainty of evidence was assessed by using the grading of recommendation assessment, development and evaluation framework (GRADE) [46]. The assessed domains were suggested by the GRADE handbook [47], which includes number of studies, risk of bias, inconsistency, indirectness, imprecision, publication bias, number of studies, and intervention effects (SMD and 95% CI). Two authors (as indicated above) independently completed the assessment and a consensus was reached by discussion when necessary. The GRADE assessment results are presented in Table S8.

2.9. Network meta-analysis

Among the included RCTs, a Bayesian network meta-analysis (NMA) was further applied to 28 parallel-arm, placebo/usual care/sham-controlled RCTs, which allowed the simultaneous comparison of multiple arms of the RCTs, including direct and indirect comparisons. Bayesian NMA was conducted using the web-based tool MetaInsight v.6.4.0, which enables users to interactively conduct NMA for both binary and continuous outcomes for fixed and random effects models based on R packages and codes [48]. The analyses were applied to each comparison twice as the comparisons were categorized by intervention type or specified intervention.

The Bayesian NMA procedures are consistent with the literature with respect to the following steps [49]: First, the network geometry plot consisted of nodes and edges that represented the number of studies that directly examined a treatment and compared two given treatments. Second, a random-effects model based on the Markov chain Monte Carlo (MCMC) simulation method was built to obtain the pooled effect sizes using the Bayesian consistency model. Subsequently, the relative effect

Table 2
Results of effect sizes and heterogeneity obtained in the meta-analyses.

Variables	Based on post-test outcomes					Based on mean change between pre-/post-tests				
	k	SMD	95% CI	I^2	$P_{subgroup}$	k	SMD	95% CI	I^2	$P_{subgroup}$
Parallel-RCTs										
Total sleeping time (TST, objective-measured)	10	0.00	[-0.25, 0.26]	58%*	0.340	11	0.12	[-0.14, 0.39]	62%**	0.050
Sleeping pills	4	0.22	[-0.03, 0.47]†	14%		4	0.11	[-0.11, 0.34]	0%	
ADHD drugs	2	−0.45	[-2.26, 1.36]	92%***		3	−0.07	[-1.22, 1.08]	85%**	
BS intervention	1	−0.06	[-0.48, 0.36]	-		1	−0.27	[-0.69, 0.15]	-	
Light therapy	1	−0.46	[-1.11, 0.18]	-		1	0.31	[-0.33, 0.95]	-	
CPA	1	0.06	[-0.38, 0.50]	-		1	0.72	[0.26, 1.18]**	-	
Environmental technique	1	−0.21	[-0.73, 0.31]	-		1	−0.08	[-0.60, 0.44]	-	
Sleeping duration (self-reported)	5	−0.07	[-0.25, 0.11]	0%	0.470	4	−0.34	[-0.59, −0.08]**	0%	-
ADHD drugs	1	0.00	[-0.26, 0.26]	-		-	-	-	-	
BS interventions	4	−0.13	[-0.38, 0.12]	0%		4	−0.34	[-0.59, −0.08]**	0%	
Sleeping efficacy (SE)	9	0.40	[-0.01, 0.81]†	88%***	0.490	7	0.24	[-0.02, 0.50]†	53%†	0.070
Sleeping pills	4	0.72	[-0.20, 1.64]	94%***		3	0.31	[-0.04, 0.66]†	51%	
ADHD drugs	2	0.23	[-0.33, 0.79]	69%†		1	0.60	[0.02, 1.18]*	-	
BS intervention	1	0.00	[-0.42, 0.42]	-		1	−0.33	[-0.75, 0.10]	-	
Light therapy	1	−0.14	[-0.78, 0.49]	-		1	0.18	[-0.46, 0.82]	-	
CPA	1	0.33	[-0.12, 0.77]	-		1	0.36	[-0.09, 0.81]	-	
Sleep onset latency (SOL)	13	−0.38	[-0.58, −0.18]***	55%**	0.570	12	−0.41	[-0.67, −0.14]**	67%***	0.300
Sleeping pills	3	−0.74	[-1.36, −0.13]*	67%†		3	−0.68	[-1.08, −0.27]**	33%	
ADHD drugs	2	−0.17	[-0.43, 0.09]	6%		1	−0.39	[-0.96, 0.18]	-	
BS intervention	5	−0.28	[-0.65, 0.08]	64%*		5	−0.36	[-0.88, 0.16]	82%***	
Light therapy	1	−0.31	[-0.95, 0.32]	-		1	−0.23	[-0.86, 0.41]	-	
CPA	1	−0.52	[-0.98, −0.07]*	-		1	−0.63	[-1.09, −0.18]**	-	
Environmental technique	1	−0.31	[-0.83, 0.21]	-		1	0.06	[-0.46, 0.58]	-	
Sleep disturbance	32	−0.27	[-0.43, −0.11]**	79%***	0.020	29	−0.30	[-0.42, −0.17]***	61%***	0.480
Sleeping pills	8	−0.30	[-0.84, 0.23]	93%***		7	−0.10	[-0.41, 0.22]	76%***	
ADHD drugs	8	−0.10	[-0.28, 0.09]	48%		6	−0.25	[-0.52, 0.01]†	45%	
BS interventions	10	−0.44	[-0.62, −0.27]***	44%†		10	−0.48	[-0.70, −0.27]***	63%**	
Neurofeedback	1	0.13	[-0.93, 1.19]	-		1	−0.53	[-1.62, 0.55]	-	
Light therapy	1	0.54	[-0.10, 1.19]	-		1	0.08	[-0.56, 0.71]	-	
CPA	1	−0.56	[-1.01, −0.10]	-		1	−0.22	[-0.66, 0.23]	-	
Environmental technique	1	0.08	[-0.44, 0.60]	-		1	−0.04	[-0.56, 0.48]	-	
Nutrient supplement	1	−0.09	[-0.72, 0.55]	-		1	−0.29	[-0.93, 0.35]	-	
BS interventions + ADHD drugs	1	−0.36	[-0.69, −0.04]*	-		1	−0.40	[-0.73, −0.08]*	-	
Latency to persistent sleep (LPS)	4	0.01	[-0.21, 0.22]	35%	0.280	4	0.10	[-0.06, 0.26]	0%	0.620
Sleeping pills	1	−0.16	[-0.50, 0.18]	-		1	0.02	[-0.31, 0.36]	-	
ADHD drugs	3	0.08	[-0.20, 0.35]	43%		3	0.12	[-0.06, 0.31]	0%	
Daytime sleepiness (all BS interventions)	6	−0.17	[-0.36, 0.03]†	22%	-	6	−0.29	[-0.51, −0.06]*	40%	-
Crossover-RCTs										
Total sleeping time (TST, objective-measured)	7	−0.31	[-0.72, 0.09]	80%***	0.090	Due to the insufficient data, SMD of mean change is unavailable for cross-over RCTs				
ADHD drugs	5	−0.49	[-1.01, 0.02]†	81%***						
Neuro-stimulant	1	0.08	[-0.66, 0.82]	-						
Environmental technique	1	0.16	[-0.12, 0.45]	-						
Sleeping duration (self-reported, all ADHD drugs)	3	−0.40	[-0.74, −0.07]*	0%	-					
Sleeping efficacy (SE)	6	−0.16	[-0.41, 0.09]	34%	0.040					
ADHD drugs	4	−0.35	[-0.61, −0.08]**	0%						
Neuro-stimulant	1	0.00	[-0.74, 0.74]	-						
Environmental technique	1	0.15	[-0.14, 0.44]	-						
Sleep onset latency (SOL)	7	0.40	[-0.05, 0.84]†	82%***	<0.001					
Sleeping pills	1	−0.58	[-1.13, −0.04]*	-						
ADHD drugs	5	0.70	[0.33, 1.07]***	55%						
Environmental technique	1	−0.06	[-0.35, 0.23]	-						

(*continued on next page*)

Table 2 (*continued*)

Variables	Based on post-test outcomes					Based on mean change between pre-/post-tests				
	k	SMD	95% CI	I^2	$P_{subgroup}$	k	SMD	95% CI	I^2	$P_{subgroup}$
Sleep disturbance	5	−0.02	[-0.31, 0.27]	55%	0.670					
ADHD drugs	4	−0.01	[-0.41, 0.39]	60%						
Environmental technique	1	−0.12	[-0.40, 0.17]	-						

Note. †0.05≤p< 0.10, *p< 0.05, **p< 0.01, and ***p < 0.001. SMD denotes standardized mean difference, which represents Hedges' adjusted *g* in RevMan 5.4.1 and includes an adjustment of small sample bias. Non-pharm denotes non-pharmaceutical interventions. RCT denotes randomized controlled trial.

size of each evaluated outcome was calculated using the mean difference (MD) of the continuous variables. The goodness-of-fit of the model was assessed using residual deviance (i.e., deviance information criterion [DIC]).

The ranking of the evaluated interventions was completed through MCMC simulations by performing 20,000 sample iterations [50]. The ranking results were obtained using Litmus Rank-O-Gram plots [51]. In these plots, the higher surface under the cumulative ranking curve (SUCRA) values and cumulative ranking curves near the top left indicated a probable better performance of the intervention.

3. Results

3.1. Overview of included studies

A PRISMA flow diagram describing the search process is shown in Fig. 1. Forty papers published between 1995 and 2025 were included in the meta-analysis. The background information of the included studies is shown in Table S5. A total of 4361 children and/or adolescents with ADHD aged 2–17 years were included in the reviewed RCTs (particularly, for parallel-RCTs: total n = 3902, intervention n = 2384, control n = 1518; for cross-over RCTs, total n = 459). Apart from four studies with no disclosure of the sex of the participants, 41.7–100% were male. Among the studies with reports of prior status on medication, six studies included drug-naïve participants, two studies included only treatment-naïve participants, and in other studies, 29.0–91.2% of the participants were on prior medication. Furthermore, the reviewed RCTs were conducted in 13 countries or districts including USA (11 RCTs), Australia (5 RCTs), Canada (4 RCTs), Iran (4 RCTs), the Netherlands (4 RCTs), Brazil (2 RCTs), and 1 RCT was performed in each of the following countries/districts: Belgium, Egypt, Germany, Hong Kong (China), Spain, Sweden, and Turkey. Moreover, among the 40 included RCTs, 30 were parallel-arm RCTs and 10 were crossover-designed RCTs.

3.2. Risk of bias and study quality

Based on the assessment of RoB 2 (Table S4), 75% (30 studies) of the included RCTs showed a low risk of bias, whereas another 10 RCTs were considered to have some concerns. Potential sources of bias included deviations from the intended interventions (e.g., limitations in blinding), a considerable proportion of missing outcome data, and potential selection bias in reporting the results.

3.3. Design of evaluated interventions

Among the included RCTs, most used a placebo, usual care, or sham in the control group, whereas two studies used active controls [52,53]. Regarding the intervention types, (i) 25 studies applied pharmaceutical intervention which comprising 8 studies of sleeping pills (i.e., melatonin, zolpidem, eszopiclone, L-dopa, L-theanine, and cyproheptadine) and 17 studies of ADHD drugs (i.e., stimulant and non-stimulants); (ii) 16 studies examined non-pharmaceutical interventions (i.e., behavioral sleep [BS] interventions, neurofeedback, cognitive training-combined/engaged physical activity [CPA], light therapy, environmental technique, and slow oscillating transcranial direct current stimulation [so-tDCS]); (iii) 1 study examined the effect of nutrient supplement (i.e., probiotic [54]); and (iv) 1 study tested the efficacy of a combined intervention protocol of BS interventions and a stimulant drug [55].

Most of the included studies tested the intervention effect in a chronic manner (i.e., two weeks to four months), whereas only one study used a single-bout intervention [56]. Among the RCTs incorporating an arm of a training program, the programs lasted for 3–14 weeks, had a frequency of 1–3 times per week, and a session duration of 45 min to 3 h.

3.4. Results of PWMA

3.4.1. Sleep duration

Among the parallel-arm RCTs, the reviewed interventions had no effect on the ***objective-measured TST*** of participants when compared with those of the control group (post-test outcomes: intervention n = 376 and control n = 295, SMD: 0.00, 95% CI [−0.25, 0.26], $p_{overall}$ = 0.97; I^2 = 58%, $p_{heterogeneity}$ = 0.01; $p_{subgroup}$ = 0.34; mean change: intervention n = 384 and control n = 311, SMD: 0.12, 95% CI [−0.14, 0.39], $p_{overall}$ = 0.36; I^2 = 62%, $p_{heterogeneity}$ = 0.003; $p_{subgroup}$ = 0.05). Among the crossover-arm RCTs, no significant effect was observed for the TST of intervention participants, when compared with those of the control group (post-test outcomes: intervention n = 276 and control n = 273, SMD: −0.31, 95% CI [−0.72, 0.09], $p_{overall}$ = 0.13; I^2 = 80%, $p_{heterogeneity}$<0.001; $p_{subgroup}$ = 0.09). Based on the self-reported data of other parallel-arm RCTs, no significant effect on ***subjective sleep duration*** of participants was found in post-test outcomes of the intervention groups in comparison with that of controls (intervention n = 407 and control n = 193, SMD: −0.07, 95% CI [−0.25, 0.11], $p_{overall}$ = 0.45; I^2 = 0%, $p_{heterogeneity}$ = 0.87; $p_{subgroup}$ = 0.47). However, a significant small effect was indicated by the mean change comparisons in BS interventions (intervention n = 124 and control n = 124, SMD: −0.34, 95% CI [−0.59, −0.08], $p_{overall}$ = 0.009; I^2 = 0%, $p_{heterogeneity}$ = 0.91). Furthermore, a significant small effect of reducing subjective sleep duration was found in the ADHD drugs reported by crossover-arm RCTs, when compared with that of the control groups (post-test outcomes: intervention n = 95 and control n = 58, SMD: −0.40, 95% CI [−0.74, −0.07], $p_{overall}$ = 0.02; I^2 = 0%, $p_{heterogeneity}$ = 0.59).

3.4.2. Sleeping efficacy

Compared with the control group, a marginally non-significant increase in SE was found in participants subjected to interventions among the parallel-arm RCTs (post-test outcomes: intervention n = 667 and control n = 363, SMD: 0.40, 95% CI [−0.01, 0.81], p = 0.06; I^2 = 88%, p < 0.001; $p_{subgroup}$ = 0.49; mean change: intervention n = 339 and control n = 247, SMD: 0.24, 95% CI [−0.02, 0.50], $p_{overall}$ = 0.07; I^2 = 53%, $p_{heterogeneity}$ = 0.05, $p_{subgroup}$ = 0.07). Among the crossover-arm RCTs, no significant effect was found in the intervention groups compared with the controls (post-test outcomes: intervention n = 218 and control n = 218, SMD: −0.16, 95% CI [−0.41, 0.09], $p_{overall}$ = 0.21; I^2 = 34%, $p_{heterogeneity}$ = 0.18; $p_{subgroup}$ = 0.040).

3.4.3. Sleep onset latency

An overall significant small effect of reducing SOL was demonstrated for the interventions examined in the parallel-arm RCTs, when compared with that of the control groups (post-test outcomes: intervention n = 665 and control n = 423, SMD: −0.38, 95% CI [−0.58,

−0.18], $p < 0.001$; $I^2 = 55\%$, $p = 0.009$; $p_{subgroup} = 0.57$; mean change: intervention n = 382 and control n = 354, SMD: −0.41, 95% CI [−0.67, −0.14], $p_{overall} = 0.003$; $I^2 = 67\%$, $p_{heterogeneity} < 0.001$, $p_{subgroup} = 0.30$). Further subgroup analysis showed that sleeping pills and CPA exhibited moderate and small effect on reducing SOL (SMDs: −0.74/-0.68 and −0.52/-0.63, respectively), whereas other interventions showed no significant effect. Based on the crossover-arm RCTs, a significant moderate effect of increasing SOL was observed in the ADHD drug groups when compared with controls (SMD: 0.70, 95% CI [0.33, 1.07], $p_{overall} < 0.001$; $I^2 = 55\%$, $p_{heterogeneity} = 0.07$).

3.4.4. Sleep disturbance

Although no significant effect was observed by the results of crossover-arm RCTs (post-test outcomes: intervention n = 255 and control n = 252, SMD: −0.02, 95% CI [−0.31, 0.27], $p = 0.89$; $I^2 = 55\%$, $p = 0.06$; $p_{subgroup} = 0.67$), an overall significant small effect of reducing sleep disturbance was found in the eight types of interventions reported by the parallel-arm RCTs (post-test outcomes: intervention n = 2274 and control n = 1412, SMD: −0.27, 95% CI [−0.43, −0.11], $p_{overall} = 0.001$; $I^2 = 79\%$, $p < 0.001$; $p_{subgroup} = 0.02$; mean change: intervention n = 1764 and control n = 1229, SMD: −0.30, 95% CI [−0.42, −0.17], $p_{overall} < 0.001$; $I^2 = 61\%$, $p_{heterogeneity} = 0.48$, $p_{subgroup} = 0.30$). Subgroup analysis suggested that BS interventions, CPA, and a combined BS-MPH protocol exhibited a significant small effect on reducing sleep disturbance at the post-test, when the intervention group was compared with the control group (SMD: −0.43, −0.56, and −0.36, respectively); for mean change comparison, only BS intervention maintained a significant small effect (SMD: −0.48). In particular, BS interventions with low risk of bias demonstrated a significant small effect on reducing sleep disturbance in participants of the intervention groups compared to control groups (SMD: −0.43, Figure S1.1-e2).

3.4.5. Latency to persistent sleep

LPS data were extracted from parallel-arm RCTs. No significant effect of LPS was observed in the evaluated interventions when they were compared with the controls (post-test outcomes: intervention n = 437 and control n = 225, SMD: 0.01, 95% CI [−0.21, 0.22], $p_{overall} = 0.96$; $I^2 = 35\%$, $p = 0.20$; $p_{subgroup} = 0.28$; mean change: intervention n = 437 and control n = 225, SMD: 0.10, 95% CI [−0.06, 0.26], $p_{overall} = 0.23$; $I^2 = 0\%$, $p_{heterogeneity} = 0.77$; $p_{subgroup} = 0.62$).

3.4.6. Daytime sleepiness

Data on daytime sleepiness were obtained from parallel-arm RCTs that assessed BS interventions. In accordance, a trivial and marginally non-significant effect on reducing daytime sleepiness was revealed in the post-test outcomes of intervention participants, when they were compared with the controls (intervention n = 294 and control n = 289, SMD: −0.17, 95% CI [−0.36, 0.03], $p = 0.09$; $I^2 = 22\%$, $p = 0.27$). However, a significant small effect was observed by the mean change comparison (intervention n = 294 and control n = 289, SMD: −0.29, 95% CI [−0.51, −0.06], $p_{overall} = 0.01$; $I^2 = 40\%$, $p_{heterogeneity} = 0.14$).

3.5. Publication bias

Publication bias of all outcomes in the PWMA was assessed using funnel plots (Fig. S2), Egger's test, and the trim-and-fill method. As shown in Table S7, no publication bias was found in Egger's test. Nonetheless, the trim-and-fill method inputted missing study(ies) into LPS and daytime sleepiness obtained from parallel-RCTs, and SE calculated based on crossover-RCTs. The estimated SMDs with 95 %CI are shown in Table S7.

3.6. Evidence grading

According to the results of GRADE assessment (Table S8), evidence obtained from crossover-RCTs prevalently showed a moderate-to-very low certainty. In contrast, the synthesized evidence emerging from parallel-RCTs exhibited moderate to high certainty. Therefore, PWMA results of this study demonstrated (1) a high certainty that the existing interventions had a significant small effect on reducing the SOL, and (2) a moderate certainty that the existing interventions had a significant small effect on reducing sleep disturbance in children/adolescents with ADHD. However, the current evidence indicates a moderate certainty that existing interventions had no significant effect on sleep duration (both objectively and subjectively measured data), SE, LPS, or daytime sleepiness.

3.7. Results of Bayesian NMA

Based on the results of SOL and sleep disturbance, a Bayesian NMA was conducted in 28 parallel-arm RCTs that used a placebo/usual care/sham control group for comparison. The relative effects (mean difference [MD] and 95% credibility interval [CrI]) of reviewed intervention types on SOL showed that sleeping pills had the largest MD than other interventions in comparisons either by post-test outcomes or mean change (Fig. S4–1). Regarding the relative effects on sleep disturbance, BS interventions had the largest MD in comparison by post-test outcomes, while it had the second largest MD in comparison by mean change (Fig. S4–2). No significant inconsistencies were observed in the results of the node-splitting analyses (all $p > 0.05$).

Moreover, based on MCMC simulations, SUCRA values also suggested probable performance ranking results of intervention types and specified interventions on SOL and sleep disturbance, as shown in Fig. 2 and Table S9. To balance the result, the summed SUCRA score of respective comparisons by post-test outcomes and mean change was used to compare. Given the SOL, the ranking of the included intervention types was sleeping pills, CPA, ADHD drugs, light therapy, BS interventions, and controls. In terms of sleep disturbance, the ranking of the reviewed intervention types was: BS interventions, a BS-MPH combined protocol, a probiotic, neurofeedback, ADHD drugs, sleeping pills, CPA, light therapy, and controls. The prior distributions of SOL and SD by intervention types are also presented in Fig. S5. This exhibited a consistent estimation on treatment performance, which may inform future studies and reduce the research bias.

4. Discussion

In all, this study suggested that (1) the existing interventions (any type) were unlikely to increase SE or sleep duration, or reduce LPS or daytime sleepiness; however, (2) these interventions (any type) were likely to significantly reduce sleep disturbance at a small level; and (3) the existing interventions (any type) significantly reduced SOL at a small effect. Moreover, among the reviewed intervention types, sleeping pills may be the most effective one to shorten SOL while BS interventions may have the best performance in reducing sleep disturbance.

Sleep problems and ADHD may have a bidirectional and complex relationship. In one hand, children and adolescents with ADHD are prevalently affected by sleep problems; in another hand, hyperactivity is significantly associated with short sleep duration with odds ratio (OR) 1.60 among children with ADHD [8]. This may attribute to not only various neurobiological mechanisms including disruptions in circadian rhythms, dopaminergic dysfunction, and increased physiological arousal level, but also negative influence of poor sleep quality on aggravation of ADHD symptoms [19,57].

In particular, methylation processes are implicated in the regulation of circadian rhythms and sleep-wake cycles. Disruptions in methylation pathways can contribute to circadian rhythm sleep disorders (CRSDs) such as delayed sleep phase disorder (DSPD) and advanced sleep phase disorder (ASPD) [58]. Genetic polymorphisms affecting methylation-related genes may influence melatonin synthesis and circadian clock gene expression, impacting sleep timing and quality [59].

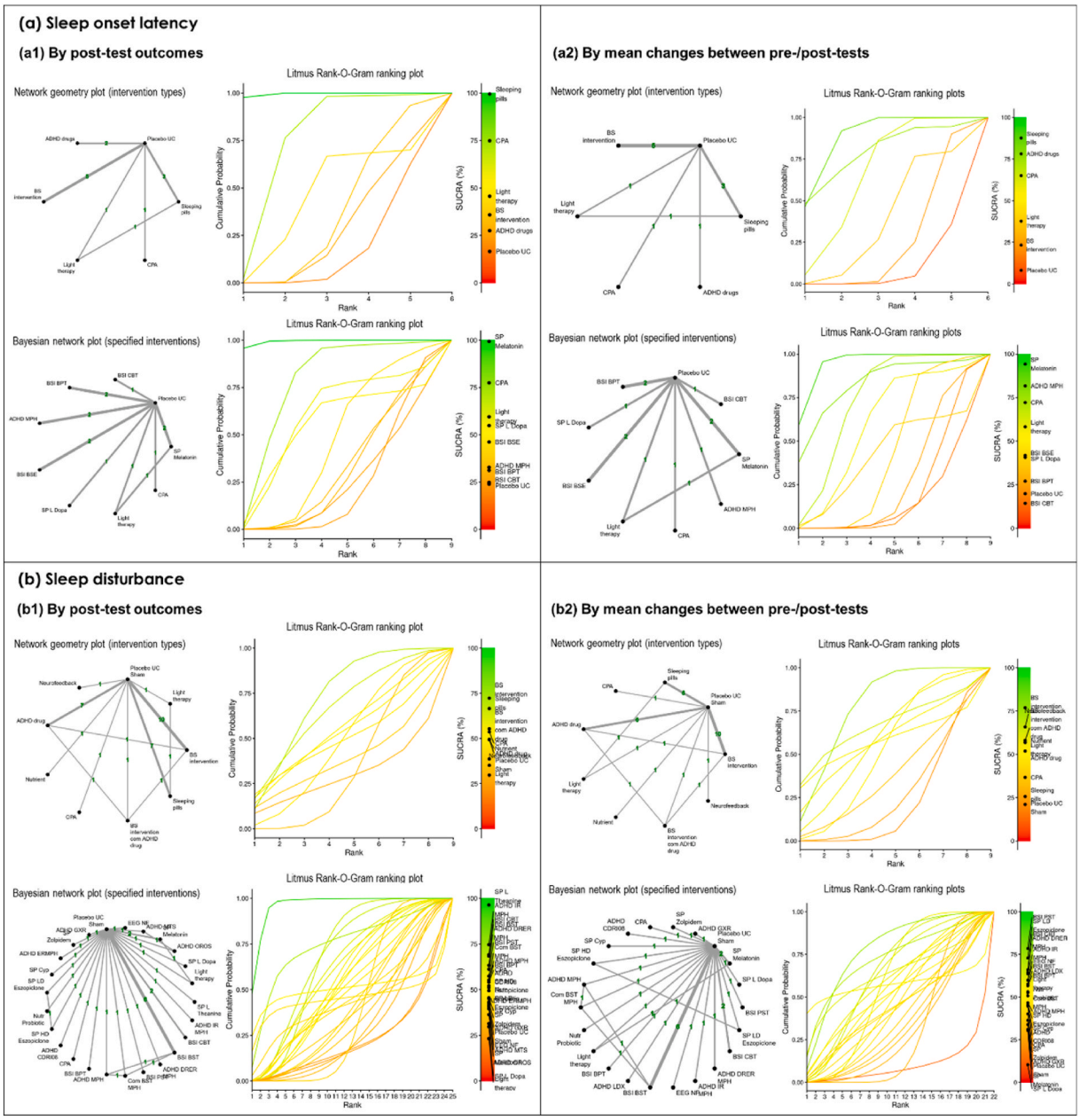


Fig. 2. Network geometry plots, Bayesian network plots, and Litmus Rank-O-Gram ranking plots of reviewed interventions.
Note. BS intervention/BSI denotes behavioral sleep intervention. ADHD denotes ADHD drugs. SP denotes sleeping pills. UC denotes usual care. MPH denotes methylphenidate. MTS denotes methylphenidate transdermal system. OROS denotes the once-daily oral form of methylphenidate (osmotic-release oral system). EEG denotes electroencephalography. NF denotes neurofeedback. GXR denotes guanfacine extended release. LDX denotes lisdexamfetamine dimesylate. Cyp denotes cyproheptadine. IR denotes immediate release. DR denotes delayed release. ER denotes extended release. HD denotes high dose. LD denotes low dose. BST denotes behavioral sleep training. PST denotes parent sleep training. CBT denotes cognitive behavioral therapy. CPA denotes cognitive training combined physical activity. BPT denotes behavioral parent training.
In network geometry plots, the size of the nodes and thickness of edges represent the number of studies that examined a treatment and compared two given treatments respectively (Owen et al., 2019).
In Bayesian network plots, numbers on the line indicate number of trials conducted for the comparison. Any shaded areas indicate existence of multi-arm trials between the comparisons (Owen et al., 2019).
Intervention ranking was completed through the introduction of Markov chain Monte Carlo simulations. More specifically, the analysis model performed 20,000 sample iterations. The ranking results are demonstrated by Litmus Rank-O-Gram plots: Higher SUCRA (Surface Under the Cumulative Ranking Curve) values and cumulative ranking curves nearer the top left indicate better performance (Nevill et al., 2023).

Furthermore, dopamine plays a key role in moderating the arousal level and REM/NREM sleep transitions. According to the previous studies [60,61], dopaminergic dysfunction with dopamine transporter (DAT) and receptors (D1/D2), as well as the associated gene expression (e.g., the dopamine transporter gene DAT1) may disturb the arousal regulation, reward sensitivity, and sleep-wake stability. Thus, imbalanced level of dopamine may jeopardize the normality of sleep-wake cycle.

Of this, the existing interventions (including pharmaceutical and non-pharmaceutical types) have been designed to help children and adolescents with ADHD to reduce their malfunctions and rehabilitate towards normalization. For pharmaceutical treatments, sleeping pills

may regulate circadian rhythms or moderate brain activity by some particular biomolecular pathways. However, the drugs for treating ADHD, such as stimulant (e.g., MPH), may overly elevate the arousal level in children and adolescents with ADHD and advocate their sleep disturbance. Among the non-pharmaceutical interventions, behavioral sleep hygiene or sleep education (e.g., cognitive-behavioral therapy, CBT) address dysfunctional beliefs about sleep with behavioral interventions (such as stimulus control and sleep restriction). These techniques focus on modifying maladaptive thoughts and behaviors related to sleep, improving sleep hygiene, and managing arousal and cognitive patterns that interfere with sleep.

In particular, for ADHD drugs, neither stimulants nor non-stimulants showed significant improvements in sleep outcomes of the intervention groups in parallel-arm RCTs. The PWMA results of this review are consistent with the literature, that is, sleep problems in children and adolescents with ADHD may be induced/aggregated by the side effects of drugs used for ADHD treatment [32], although these could vary in individual responses [62]. Nonetheless, several studies have indicated that particular formulations of stimulants (e.g., sustained/extended-release MPH) with an optimal dose may decrease the risk of side effects and may not negatively affect sleep in children or adolescents with ADHD [62–66]. Thus, personalized and optimized ADHD medications are warranted for these youth.

In terms of sleeping pills, results yielded by post-test outcomes or mean change showed an overall non-significant effect on increasing TST (k = 4, SMD: 0.22 and 0.11) with small heterogeneity, and an overall significant relatively-large effect on decreasing SOL (k = 3, SMD: −0.74 and −0.68) with small-moderate heterogeneity. In particular presented in Table S6, melatonin presented a significant large effect on reducing SOL (k = 2, SMD: −0.98 [67,68]), whereas L-dopa showed a non-significant small effect (k = 1, SMD: −0.21, [69]). Nonetheless, the post-test outcomes showed that L-theanine [70] presented a significant large effect on increasing SE (k = 1, SMD: 2.25) and reducing sleep disturbance (k = 1, SMD: −2.53). L-theanine has shown promising effects on stress reduction and mood relaxation, effects relevant to its regulation of brain state modulation via the release or reduction of brain neurotransmitters gamma aminobutyric acid (GABA), dopamine, and serotonin [71]. However, only one included study reported the post-test effects of L-theanine (with no baseline data available, [70]). Thus, more rigorous evaluation of this promising drug is warranted. For other sleeping pills targeting on sleep disturbance, both Z-drugs (k = 3) and antihistamines (k = 1) presented no effect, while melatonin presented a large heterogeneity but a non-significant trivial to small effect.

Regarding other interventions (Table S6, Figure S1, and Figure S3), with respect to SOL in particular, BS interventions showed no obvious effect in three studies with low risk of bias, whereas CPA presented a significant moderate effect on reducing SOL. In contrast, BS interventions exhibited a significant effect on sleep disturbance (i.e., six studies with low risk of bias, SMD: −0.43 by post-test outcomes and −0.41 by mean change, close to moderate effect). This result is in line with that of a previous meta-analysis [9], which reported a moderate certainty of evidence for behavioral interventions to reduce sleep disturbance in children with ADHD to a moderate level (SMD: −0.49). However, the consistent comparison results by post-test outcomes and mean change showed that neurofeedback, CPA, light therapy, an environmental technique, and a nutrient supplement had no effect on sleep disturbance.

In line with the results of PWMA, Bayesian NMA results suggested that sleeping pills had the best performance in achieving reduced SOL, whereas BS intervention was assessed to be the most effective one in decreasing sleep disturbance. In particular, with consideration of both post-test outcome and mean change (Table S9), melatonin showed the probable best performance to reduce SOL, and PST (one of BS interventions) was estimate to be the best in decreasing sleep disturbance. Thus, these interventions may be promising for adjusting the targeted sleep outcomes of children and adolescents with ADHD.

Furthermore, in this review, majority of included studies had no specification on exclusion of children and adolescents with ADHD who had comorbid sleep disorders (e.g., RLS, PLMS, OSA). According to the four studies which included participants with primary insomnia only (Fig. S6), CBT intervention presented a small to moderate effect on reducing sleep disturbance (SMD by post-test outcomes: −0.42, p = 0.05; SMD by mean change: −0.50, p = 0.02 [72]), whereas L-dopa [69], Zolpidem [73], and IR-MPH [74] exhibited no significant effect (p > 0.05). This finding is consistent to the study reported by England et al. (2011) [69], i.e., L-dopa exhibited a significant effect on patients with RLS/PLMS rather than those with ADHD-only. This may imply that: (1) the comorbid sleep disorders may largely contribute to the sleep problems of children and adolescents with ADHD; (2) the existing intervention (especially pharmaceutical interventions) may be more effective to treat these comorbidities rather than the primary insomnia; and (3) specific CBT protocols for ADHD-related sleep issues are less established than for other groups with primary insomnia, although CBT (i.e., CBT-I) is widely recognized as the first-line, gold standard treatment for chronic insomnia disorder [75]. This is in line with the findings of a previous review [76], which found inconsistent results of CBT technique and difficulty in long-term maintenance of intervention effect in individuals with neurodevelopmental conditions. However, more evidence is warranted to examine this inconclusive point.

Overall, the current review implied that both pharmaceutical and non-pharmaceutical interventions may play the important but different roles in reducing sleep problems of children and adolescents with ADHD. Future intervention designs may integrate the effective options into a multi-component strategy. This is possible to achieve a combined or synergetic effect with optimized pharmaceutical dose and intervention regimen, which may concurrently increase the intervention safety and participants' adherence. Nonetheless, due to the large diverse in individual situation and condition, a tailored or personalized intervention strategy is also highly warranted for children and adolescents with ADHD.

This study had some limitations that should be acknowledged. First, results should be interpreted with caution owing to limitations regarding the study design, unbalanced participants' characteristics, intervention design, and reporting bias in the reviewed studies. The acknowledged limitations of the study designs in the included RCTs comprise small sample size, concerns with blinding, adherence issues with objective measures, unbalanced baseline, lack of placebo- or sham-controls, unclear carryover effect in crossover-arm RCT, lack of follow-up, insufficient data or measures, attrition or missing data, and invalidity of tools or measures. Regarding participants' characteristics, homogeneity in age range, sex distribution, comorbidity, and/or recruitment location was attributed to limited generalization and failure of moderator detection. In contrast, diverse ADHD symptoms or medication responses, changed medication plans during study (e.g., participants took melatonin or self-administered stimulants), as well as inclusion of comorbid sleep disorders (such as RLS, PLMS, or OSA), also contributed to the study limitations. Furthermore, the included studies reported limitations regarding the intervention design, including the timing of the intervention, short intervention duration, inevitable confounders, and the need for a tailored/personalized design. Moreover, parent- or clinician-reported data may have been influenced by expectations of interventions, resulting in social desirability bias. Finally, the studies selected for review were also limited because only those with specific terms mentioned in the title or abstract were screened for further analysis, and studies published in languages other than English or in conference abstracts, government reports, textbooks, or unpublished dissertations were excluded.

5. Conclusions

By employing PWMA and Bayesian NMA, the current study systematically reviewed and analyzed the relevant RCTs to compare the

effects of interventions and controls on sleep among the targeted population. According to the PWMA results, the existing interventions of any type likely significantly improve the quality of sleep of participants by shortening SOL and reducing sleep disturbance. However, SE, sleep duration, LPS, or daytime sleepiness were unlikely changed by the existing interventions. Moreover, the NMA results suggest that sleeping pills and BS interventions are probably to reduce SOL and sleep disturbance, respectively. More evidence from rigorous assessment of these promising options is warranted to confirm this effect, especially in consideration of those with or without comorbid sleep disorders. Owing to the diversity of sleep issues, a multi-component and tailored/personalized sleep interventions are crucial for young individuals with ADHD.

Author contribution

YF, ZW and FS contributed to the study design. ZW, MZ and FS provided professional consultation. YF and ZW conducted the data collection and analysis. YF, ZW and FS wrote the manuscript. All authors reviewed the manuscript critically and significantly contributed to the intellectual content of the manuscript.

Data availability statement

The data that support the findings of this study are available from the corresponding author (Dr. Fenghua SUN), upon reasonable request.

Ethic statement

Since this review was conducted and analyzed based on published data, Internal Review Board (IRB) approval was not required.

Declaration of generative AI and AI-assisted technologies in the writing process

None.

Conflict of interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.smrv.2026.102303.

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